

Disk-Completion Kernels

Optimal Single-Exponential Representative Families for Surface Frontiers

A certified search compression toward quasi-polynomial unknot recognition

OpenAI

Research continuation prepared for the ProveIt project

October 9, 2026

Abstract

A frontier can have Bell-many distinct connectivity states without requiring Bell-many representatives for a particular terminal question. We study the question relevant to a compressing-disc search: which partial surfaces can be completed to one disk by gluing along specified, disjoint boundary arcs? For compact surfaces with boundary, the defect $\delta(S) = \text{comp}(S) - \chi(S)$ satisfies an exact gluing identity: defects add, together with the cycle rank of the component-attachment graph. Thus a disk completion requires disk pieces and a tree attachment graph, with no orientability hypothesis.

For r labelled arcs, the resulting disk-completion matrix has exact rank 2^{r-1} over $\mathbb{F}_2$. Its block with c components has exact rank $\binom{r-1}{c-1}$. An explicit root-transversal factorization gives a deterministic, minimum-weight representative *subset* of these sizes, preserving every disk-completion query. The sizes are sharp even for unrestricted representative subsets. We prove a typed composition theorem, a 3^b bound when active labels vary inside a pool of size b , and a signed-connectivity extension. A supplied finite surface-assembly grammar can consequently be searched in $2^{O(b)}$ times its encoded polynomial overhead, instead of enumerating all partitions. Under a complete polynomial-size geometric grammar and $b = O((\log n)^2)$, this would give a quasi-polynomial recognition route.

The delivery includes a weighted reducer, an independent algebraic certificate checker, a complete finite-grammar solver, and an independent triangulated-surface replay. Thirty-six test methods pass; additional audits check 813,297 direct compatibility-matrix entries, 1,000 weighted families, 1,000 literal surface gluings, and 200 complete grammars. Paired abstract-grammar experiments show 3.98–11.91 times lower median running times, while a one-query control is slower after compression. These are not timings of the maintained knot recognizer. The general rank-based method is classical; the contribution is its sharp disk-specific formulation, topological contract, and checked implementation. No complete bounded-width grammar for all knot exteriors is established here.

Status. Proved local and supplied-grammar theorems; tested standalone code. No proof-assistant verification, native recognizer integration, or general quasi-polynomial unknot-recognition bound is claimed.

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1 Research position and the precise improvement

1.1 The current project already has a geometric source bridge

This article continues `Topology/UnknotRecognition` in ProveIt. The reviewed repository revision is

3a90fb34146c915328ab8eac6250cc2514f74ed0.

At this revision, the maintained material includes source-checked diagram-to-exterior construction and a source-bound normal-surface component census [6, 7]. In particular, the canonical pulling construction uses $20 \max(1, n)$ tetrahedra for an n -crossing validated planar diagram. A supplied normal surface can be decomposed and tested for compressing-disc components using Euler characteristic and two torus-boundary parities. These are important advances: it would be incorrect to describe the entire diagram-to-manifold provenance step as still absent.

The remaining search problem is different. Fast verification of one supplied normal vector does not construct a complete, efficiently searchable collection of vectors or hierarchy moves. The present work addresses one source of search growth, namely the number of connectivity alternatives carried through a surface frontier. It neither replaces the existing compressed orbit engine nor reproves its weighted census; the classical compressed-orbit context is supplied by Agol, Hass, and Thurston [9].

The predecessor *Certified Port Quotients and Guarded Interval Attachments* [8] gives a sharp $\text{Bell}(r + 1)$ state bound for a fixed-source coning model. Those states are physically distinguishable by suitable subsequent operations. It asks whether useful restricted continuations permit $2^{O(r)}$ rather than $2^{O(r \log r)}$ search. We obtain such a bound for a *different, explicitly narrower observation*: minimum-cost completion to a disk along genuine boundary arcs. We do not identify all physical states, and we do not preserve arbitrary coning predicates.

1.2 What changes, and what does not

A partition of r labels says which frontier arcs lie on the same partial disk. An exact state table can contain $\text{Bell}(r)$ partitions. Our reducer retains at most 2^{r-1} *actual candidates* of one geometric type, while preserving the optimum for every possible disk-union cap on those arcs. A discarded candidate need not have the same behavior as any single retained candidate. Instead, for each cap that completes it, at least one no-more-expensive retained candidate completes that cap. This is the critical logical distinction between state equivalence and representative families.

The main steps are a topological reduction to a tree, an exact matrix factorization, a cost-ordered basis argument, and a replacement proof inside a finite assembly grammar. The first three do not assume a special knot class. The global application does require that a valid geometric producer expose its partial surfaces through the stated interface. We give counterexamples showing why dropping this requirement is unsound.

1.3 Relation to established methods and the novelty boundary

Deterministic rank-based compression of connectivity dynamic programs, including weighted variants, is established by Bodlaender, Cygan, Kratsch, and Nederlof [1]. Representative families for linear matroids and their exterior-algebra constructions are also classical algorithmic tools; Fomin, Lokshtanov, Panolan, and Saurabh give systematic algorithms and applications [2, 3]. Our reduced incidence minors are a concrete graphic-matroid instance of that machinery.

We do not claim the greedy basis principle, Gaussian elimination, or single-exponential connectivity dynamic programming as new inventions.

The disk-specific completion matrix is not the unrestricted “union is a forest” compatibility matrix studied in work on the Tutte polynomial [4]. We require a *connected tree*, work over $\mathbb{F}_2$, and keep component count as a grading. The explicit root-transversal coordinates make the exact ranks and matching lower bounds transparent. The research contribution here is the combined disk-topology statement, sharp completion kernel, typed replacement theorem, safety analysis, and reproducible project-facing implementation. We establish these assertions in full below; we have not established worldwide priority for every equivalent algebraic formulation.

Lackenby’s hierarchy paper is relevant geometric context, not a complexity theorem we silently assume. Its Section 9 explicitly distinguishes a bound on certain algorithmic steps from a running-time estimate for all operations [5]. Our conditional corollary will name construction, type-count, width, and bit-size obligations separately.

2 Surfaces, interfaces, and topological defect

2.1 The geometric interface contract

All surfaces are finite compact PL surfaces, possibly disconnected and nonorientable. Every connected component has nonempty boundary. An attachment arc is a subspace homeomorphic to a closed interval properly contained in a boundary circle. In a single surface, different attachment arcs are disjoint *including their endpoints*. They have disjoint collars, and gluing pairs them by homeomorphisms; corners are then smoothed. No other points are identified. These conditions ensure that the quotient is again a compact surface with boundary. Boundary intervals not used for this attachment remain available for later stages.

One label denotes one connected geometric arc. It does not denote a union of separated arcs, an entire point-incidence port, or a binary-compressed bundle of many parallel arcs. Translating the repository’s compressed incidence data into this interface requires an additional geometric theorem. The same word “port” is not such a theorem.

Write $c(S) = \text{comp}(S)$ and define

$$\delta(S) = c(S) - \chi(S).$$

For a connected orientable surface of genus g with $h \geq 1$ boundary circles, $\delta = 2g + h - 1$. For a connected nonorientable surface with $k \geq 1$ crosscaps and $h \geq 1$ boundary circles, $\delta = k + h - 1$. Thus $\delta(S) \geq 0$, and $\delta(S) = 0$ precisely when every component of S is a disk. These formulas may also be read as the elementary classification-of-surfaces test used in the following proof.

Theorem 2.1 (Exact defect under boundary-arc gluing). *Let S and T satisfy the preceding contract, and glue r pairs of boundary arcs between them. Let G be the bipartite multigraph whose vertices are the connected components of S and T , and whose edges are the r glued pairs. Include isolated vertices when a component has no arc used in this particular attachment. For $W = S \cup_{\text{arcs}} T$,*

$$c(W) = c(G), \quad \chi(W) = \chi(S) + \chi(T) - r, \quad (1)$$

$$\delta(W) = \delta(S) + \delta(T) + \beta_1(G), \quad \beta_1(G) = r - c(S) - c(T) + c(G). \quad (2)$$

In particular, W is a union of disks if and only if S and T are unions of disks and G is a forest. It is one disk if and only if the pieces are unions of disks and G is a tree.

Proof. A path in the quotient moves within connected components of the original surfaces and crosses glued intervals. Such a path determines a walk in G . Conversely, an edge of G joins the indicated surface components in the quotient. This proves the component identity, including isolated vertices.

Choose compatible triangulations in which each gluing interval is a subcomplex. Euler-characteristic inclusion–exclusion subtracts one for each glued interval, because an interval has Euler characteristic one. The intervals are pairwise disjoint, so no additional multiple-intersection terms occur. This gives the second identity in (1). Substitute it into $c(W) - \chi(W)$ and add and subtract $c(S) + c(T)$ to obtain (2). The expression for $\beta_1(G)$ is the usual edge–vertex–component formula for a multigraph, and is nonnegative.

There is free boundary in each component: finitely many pairwise disjoint proper closed intervals cannot cover a boundary circle, and a free boundary point survives gluing. Thus the nonnegativity and disk characterization of δ apply to W as well as to S, T . Every term on the right of (2) is nonnegative. Their sum vanishes exactly when both inputs are disk unions and G has no cycle. Adding connectedness gives the final assertion. $\square$

Corollary 2.2 (Defect cannot be repaired). *During a sequence of attachments satisfying the contract, an existing positive surface defect cannot be removed by attaching more surfaces with boundary. Every partial surface on a successful disk assembly is therefore a union of disks. A partial component with no remaining attachment arc cannot become connected to later material.*

Proof. Apply (2) successively. For the second statement, a component with no remaining arc is an isolated vertex in every future attachment graph. If additional material must be part of the same final disk, such a component prevents success. $\square$

A component that has lost its last live arc can still be a *terminal* disk when no other material remains. It must be handled by a separate terminal rule, not silently deleted from an intermediate partition. The delivered layered grammar always has a nonempty outgoing interface until its final cap, so its intermediate rejection of sealed components is safe.

2.2 Orientation and parallel-arc examples

No orientation signs are needed to recognize the tree case. Inductively remove a leaf of the attachment tree: it is a disk glued along one boundary interval to the rest. This attachment makes a disk boundary connected sum, irrespective of the chosen homeomorphism of the interval. One may choose the orientation of the leaf after the orientation of the remaining component is fixed.

By contrast, two disks glued along two separated intervals have $\chi = 1 + 1 - 2 = 0$ and a connected attachment graph with one cycle. Suitable gluing maps give an annulus; other choices give a Möbius band. Neither can later be made a disk by the permitted operations. Collapsing those two intervals to one abstract port would falsely predict a disk. The cycle term is not a cosmetic correction: it is precisely the information a parallel-arc compression must preserve.

3 The disk-completion matrix

3.1 Partitions and the completion predicate

Fix $r \geq 1$ and $U = \{0, 1, \dots, r-1\}$, with distinguished label 0. Let $\text{Part}(U)$ be its set of partitions. A live disk-union frontier has one labelled arc for each element of U , and every disk component meets at least one arc. It determines $p \in \text{Part}(U)$: labels lie in one block exactly when their arcs lie on the same disk. Let $|p|$ denote the number of blocks.

For $p, q \in \text{Part}(U)$, form the bipartite graph $G(p, q)$ with one vertex for each block of p or q , and an edge labelled u joining the two blocks containing u . Let $p \vee q$ be the least common coarsening, whose blocks are the connected classes generated by the two equivalence relations. Write $\mathbf{1}_U = \{U\}$ for the one-block partition. Define the zero-one matrix

$$D_r(p, q) = [p \vee q = \mathbf{1}_U \text{ and } |p| + |q| = r + 1]. \quad (3)$$

Every vertex of $G(p, q)$ has an incident edge. It is connected exactly when $p \vee q = \mathbf{1}_U$. Its vertex count is $|p| + |q|$ and its edge count is r . Consequently (3) is exactly the indicator that $G(p, q)$ is a tree. By [theorem 2.1](#), it is exactly the topological disk-completion predicate.

This interpretation depends only on which disks carry the labelled intervals. It does not require a common cyclic order on all the intervals, noncrossing partitions, or a planar embedding of $G(p, q)$. Any restrictions imposed by an ambient 3-manifold enter through the set of permitted candidates and caps and their geometric types.

3.2 Root transversals

Let

$$\mathcal{R} = \{R \subseteq U : 0 \in R\}, \quad d = |\mathcal{R}| = 2^{r-1}.$$

A set R is a transversal of p if it contains exactly one element of each block. Define

$$C(p, R) = [R \text{ is a transversal of } p], \quad R^* = \{0\} \cup (U \setminus R). \quad (4)$$

The map $R \mapsto R^*$ is an involution of $\mathcal{R}$. Let J be its permutation matrix. The vector $C(p, \cdot)$ is supported on sets of size $|p|$.

Theorem 3.1 (Exact root-transversal factorization). *Over $\mathbb{F}_2$,*

$$D_r = CJC^\top. \quad (5)$$

Equivalently, the parity of the number of sets $R \ni 0$ such that R is a transversal of p and R^ is a transversal of q is one precisely when the two disk unions complete to a disk.*

We give a minors proof, rather than assuming an unproved rule about gluing arbitrary partition encodings.

Lemma 3.2 (Rooted forest minor). *Let F be a forest on U with c components, and let B_F be an oriented vertex-edge incidence matrix over $\mathbb{F}_2$ with the row for vertex 0 removed. If $A \subseteq U \setminus \{0\}$ has size $r - c$, then the square minor $B_F[A, *]$ has determinant one if and only if $U \setminus A$ contains exactly one vertex from each component of F .*

Proof. A tree component with v vertices contributes $v - 1$ columns. Its full incidence matrix has rank $v - 1$: a leaf elimination proves that deleting any one row gives a square matrix of determinant one. If no row is deleted from some component, the selected rows of that

component contain a dependence. If two or more rows are deleted from some component, fewer than $v - 1$ selected rows remain in that component, so its columns cannot be independent. Thus a nonsingular global square minor requires exactly one omitted row per component. Under that condition it is block diagonal, after row and column permutations, with the rooted-tree matrices just described. Each determinant is one over $\mathbb{F}_2$, proving the converse as well. $\square$

Proof of theorem 3.1. If $C(p, R)C(q, R^*)$ is nonzero, then $|p| = |R|$ and $|q| = |R^*| = r + 1 - |R|$. Hence both sides of (5) vanish when $|p| + |q| \neq r + 1$.

Assume the block counts are complementary. Choose an arbitrary tree on each block of p and of q , obtaining forests F_p, F_q on U with $r - |p|$ and $r - |q|$ edges. Keep edges from the two forests distinct, even if they have identical endpoints. The concatenated reduced incidence matrix

$$B = [B_{F_p} \mid B_{F_q}]$$

is square of size $r - 1$. Its determinant is one precisely when the resulting multigraph is a tree. The graph has $r - 1$ edges and has connected classes $p \vee q$, so this is precisely $D_r(p, q) = 1$.

Expand the determinant by the columns belonging to F_p . A term selects a row set A of size $r - |p|$ for those columns and the complementary rows for F_q . All Laplace signs disappear in characteristic two. By lemma 3.2, the first minor is $C(p, U \setminus A)$. Put $R = U \setminus A$; the omitted rows for the second forest are $\{0\} \cup A = R^*$. Its minor is $C(q, R^*)$. Summing the products gives (5). The empty 0×0 determinant has value one, covering $r = 1$ without an exception. $\square$

3.3 Exact total and graded ranks

Theorem 3.3 (Sharp rank in every component grading). *For every $r \geq 1$,*

$$\text{rank}_{\mathbb{F}_2} D_r = 2^{r-1}. \quad (6)$$

For $1 \leq c \leq r$, the submatrix with row partitions having c blocks and column partitions having $r + 1 - c$ blocks has rank

$$\binom{r-1}{c-1}. \quad (7)$$

Proof. The factorization bounds the total rank by d . The homogeneous support of a row with c blocks lies in the $\binom{r-1}{c-1}$ coordinates of size c , giving the graded upper bound.

For $R \in \mathcal{R}$, let π_R be the partition with one block $(U \setminus R) \cup \{0\}$ and singleton blocks $\{u\}$ for all $u \in R \setminus \{0\}$. A transversal containing 0 must choose 0 from the first block and must choose each of the singleton vertices. Therefore

$$C(\pi_R, \cdot) = e_R.$$

The submatrix of D_r indexed by these d partitions is the permutation matrix J . In the c -block grading, the rows indexed by $|R| = c$ and columns indexed by π_{R^*} similarly give a permutation matrix of order $\binom{r-1}{c-1}$. The lower bounds match the upper bounds. $\square$

Example 3.4 (The first nontrivial dependence). At width three, write partitions in restricted-growth notation. The partitions 001, 010, and 011 have root vectors

$$C(001) = e_{\{0,2\}}, \quad C(010) = e_{\{0,1\}}, \quad C(011) = e_{\{0,1\}} + e_{\{0,2\}}.$$

Together with 000 and 012, the five partitions span four dimensions. The dependence is not an equality of physical surfaces; it is a linear identity for their completion tests.

4 Minimum-weight representative subsets and certificates

4.1 The exact guarantee

A candidate is a tuple (p, w, ω) : a frontier partition, an integer cost written in binary, and an actual witness describing the partial surface. All candidates in one reduction call must share one geometric and boundary type. This type requirement is formalized in [section 6](#); it cannot be certified from p alone.

For a finite family $\mathcal{A}$ and a cap partition q , put

$$\text{opt}(\mathcal{A}, q) = \min\{w(a) : a \in \mathcal{A}, D_r(p_a, q) = 1\},$$

with minimum $+\infty$ for the empty set. A subset $\mathcal{A}' \subseteq \mathcal{A}$ is *minimum disk-representative* if these optima agree for every $q \in \text{Part}(U)$. Costs may be negative: finiteness of the family, not positivity, is what is needed.

Theorem 4.1 (Sharp weighted disk-completion kernel). *Every finite family of candidates on $r \geq 1$ arcs has a minimum disk-representative subset $\mathcal{A}'$ with*

$$|\mathcal{A}'| \leq 2^{r-1}, \quad |\{a \in \mathcal{A}' : |p_a| = c\}| \leq \binom{r-1}{c-1}.$$

Such a subset is found deterministically by scanning candidates in nondecreasing cost and retaining precisely those root vectors independent of the vectors retained so far. The output consists of original candidates, not formal sums of witnesses.

Proof. Fix a total order refining cost order. During Gaussian elimination retain an original candidate if and only if its vector is independent of earlier retained vectors. Every discarded vector is an $\mathbb{F}_2$ -linear combination of retained vectors whose costs are no larger. All nonzero vectors have a well-defined component grading, and the coordinate spaces of different gradings are disjoint. Hence an independent collection has at most the stated number of vectors in each grading; summing gives 2^{r-1} .

Consider any candidate a completed by q . If a is retained, nothing needs to be shown. Otherwise write

$$C(p_a, \cdot) = \sum_{b \in B_a} C(p_b, \cdot), \quad w(b) \leq w(a).$$

Pair with $JC(q, \cdot)^\top$. By [theorem 3.1](#), the left side is one. Therefore at least one summand is one. That summand comes from a retained candidate completed by q at no greater cost. Apply this observation to an optimum candidate. The reverse inequality follows because the retained family is a subset of the original family. This proves equality of optima, including the case of no completion. $\square$

This argument is deterministic. It does not conclude that an odd number of solutions exists because some randomized evaluation is nonzero. The parity identity is used to certify that a *particular successful row* has a successful retained replacement. No valid witness is replaced by an XOR of surfaces.

Theorem 4.2 (Representative-size lower bound). *The bounds in [theorem 4.1](#) cannot be reduced in general, even by a method that is not based on linear algebra, as long as it must output a subset preserving all disk-completion queries. The same holds separately in each component grading.*

Proof. Take one cost-zero candidate for each π_R of [theorem 3.3](#). The query $q = \pi_{R^*}$ completes π_R and no other member of this family, since the induced completion matrix is J . Omitting that candidate changes its query optimum from zero to $+\infty$. Thus all 2^{r-1} candidates are necessary. Restricting to $|R| = c$ gives the graded lower bound. $\square$

The lower bound concerns worst-case representative *subsets*. It is not a lower bound on the running time of every algorithm for one particular cap, nor a claim that all knot-derived frontiers realize the sharp family. A single supplied cap can be tested directly against a family in polynomial time per candidate, without any exponential-dimensional preprocessing.

4.2 A certificate that checks the discarded rows

A positive witness alone proves a feasible assembly, but not that pruning preserved the best one. The reducer therefore optionally returns the following evidence: the list of retained original indices and, for every input row a , a list B_a of retained indices such that

$$C(p_a, \cdot) = \bigoplus_{b \in B_a} C(p_b, \cdot), \quad w(b) \leq w(a) \quad (b \in B_a). \quad (8)$$

A retained row can cite itself. Distinct indices are required inside each expansion. The checker receives the original ordered partitions and costs independently of the certificate, reconstructs all root vectors, and checks every identity and cost comparison. It also checks the total and graded output-size bounds. A certificate for a modified source is not accepted merely because some digest has a plausible shape.

To verify the pruning guarantee, it is not necessary to trust the producer’s pivot order or to recompute an optimal basis. Indeed, any retained subset satisfying (8) has the required replacement property. Independence is a convenient way to *produce* a small subset, while the linear expansion and cost dominance are a simple way to *check* it.

The delivered checker is deliberately separate. The producer builds root vectors by choosing one root from each block not containing 0. The checker scans all root subsets and counts the roots in each block. The producer tests compatibility by disjoint sets; the independent direct predicate uses breadth-first connectivity and the graph’s edge–vertex count. Tests disable the producer while the checker runs and mutate source partitions, costs, indices, expansions, and schema fields. These are independent code paths, not a claim that shared mathematical mistakes are impossible.

4.3 Encoded cost, memory, and input dependence

Let L be the number of input candidates, B the maximum bit length of an integer cost, and W the total encoded witness length or the size of a shared witness DAG. Let $d = 2^{r-1}$. A transparent implementation may form each vector by scanning its d coordinates and testing transversality in $O(r)$ elementary operations. Thus $O(Lrd \text{ poly}(\log(r+1)))$ bit work is a safe construction bound.

Naive elimination uses at most d pivots per input row and XORs vectors with at most d bits. Tracking a retained-row expansion also uses at most d bits per basis vector. Hence

$$O\left(Ld^2 + Lrd \text{ poly}(\log(r+1)) + LB \log(L+1) + \text{input and witness costs}\right) \quad (9)$$

is a conservative bit bound. It does not treat a d -bit Python integer as one constant-time object. The implementation’s packed integers can improve constants, not the stated guarantee.

Witnesses may be kept by immutable indices or parent pointers; copying long witness arrays must instead be charged explicitly.

The basis and its expansions use $O(d^2)$ bits. An explicit certificate with up to d original indices per row uses $O(Ld \log(L+1))$ bits. The input itself occupies $\Omega(L)$ space, and certificates retained from every stage must also be counted. One may stream certificates or check them before discarding a stage, but that is a separate storage policy. The size bound counts *retained candidates*, not total working bits.

There is no $2^{O(r)}$ total-time theorem for an arbitrarily large explicit input list without charging its length. In particular, constructing all $\text{Bell}(r)$ partitions and then reducing them still costs at least $\text{Bell}(r)$ input operations. The single-exponential search theorem in [section 6](#) assumes candidates are built from already reduced child families and a bounded explicit rule set.

5 Active interfaces and signed connectivity

5.1 A pool of potentially active arc labels

A geometric interface may specify a pool P of b possible labels, while an individual state uses only a subset $A \subseteq P$. In general different active sets cannot be merged: a future attachment may require an arc absent from one of them. Reduce each nonempty active set separately, after relabelling one of its elements as the distinguished root.

Corollary 5.1 (The active-set sum). *For one fixed external type and a pool of b labels, the total number of representatives across nonempty active sets is at most*

$$\sum_{\emptyset \neq A \subseteq P} 2^{|A|-1} = \frac{3^b - 1}{2}. \quad (10)$$

If one additional empty-interface terminal record is needed, the bound is $(3^b + 1)/2$. These bounds apply per type; independent homology or geometry types multiply the count.

Proof. Apply [theorem 4.1](#) on each active set and sum the binomial expansion of $(1 + 2)^b$. The empty interface is a separate terminal case, not a partition of positive width to which the root-vector construction is applied. $\square$

Equation (10) does not identify arbitrary partial partitions arising in coning. The predecessor's $\text{Bell}(b+1)$ physical-state distinction remains valid. Here a table of *families* is compressed for one completion predicate under a different transition contract.

5.2 An exact signed-connectivity rank

For some extensions, signs on the interface must be retained. A signed partition is a pair (p, a) where $a \in \mathbb{F}_2^U$, with two assignments identified if they differ by flipping all signs in any whole block of p . It describes the within-component relations

$$x_u + x_v = a_u + a_v \quad (u, v \text{ in one block of } p).$$

Use one fixed framing for interface labels. Any reversal needed on the cap side is transported into its assignment before these equations are combined.

Let $\mathcal{X} = \{x \in \mathbb{F}_2^U : x_0 = 0\}$ and set

$$H((p, a), x) = [x \text{ satisfies all within-block equations of } (p, a)].$$

Define $M(s, t)$ to be one when the combined unsigned partition is connected and the combined signed equations are consistent, and zero otherwise.

Theorem 5.2 (Signed root-coloring factorization). *Over $\mathbb{F}_2$,*

$$M = HH^\top, \quad \text{rank}_{\mathbb{F}_2} M = 2^{r-1}.$$

Thus minimum-cost representatives of size at most 2^{r-1} also suffice for all connected-and-consistent signed completion queries.

Proof. The (s, t) entry of $HH^\top$ is the parity of the common root-fixed solutions. An inconsistent system has no solutions. A consistent one with k connected classes has 2^{k-1} solutions after fixing $x_0 = 0$, because each class has one free flip and the class containing 0 has its flip fixed. Its parity is one exactly when $k = 1$. This proves the factorization and the upper rank bound.

For a one-block partition and any chosen root-fixed assignment a , the vector $H((1_U, a), \cdot)$ is the unit vector at a . These 2^{r-1} signed states give an identity submatrix of M . This proves equality. The same cost-ordered basis and pairing proof as [theorem 4.1](#) gives the representative statement. $\square$

This theorem concerns connectedness and consistency, not disk topology. A connected orientable annulus satisfies an orientation consistency test but has positive defect. Euler information must not be discarded. Conversely, in the genuine disk-completion case the tree condition already permits compatible orientations, so signs are not a necessary extra exponential factor. The delivery includes signed vectors and an independent signed compatibility predicate, with exhaustive small tests. It does *not* include a general signed, higher-genus surface-assembly solver.

6 Safe composition in a finite assembly grammar

6.1 What a geometric type must guarantee

The weighted theorem is a statement about a family and every cap. To use it repeatedly inside a search, one must know that an actual remaining computation can be treated as such a cap. We make that condition explicit.

Definition 6.1 (Typed boundary-arc assembly grammar). A finite typed assembly grammar is an acyclic construction specification with the following data and requirements.

- (i) A partial object is a realized compact surface with boundary and a finite labelled live-arc interface satisfying [section 2](#). A live nonterminal object has no component without a live arc. Its state includes a geometric type, its active labels, its partition, its integer cost, and a recoverable witness.
- (ii) Local rules glue prescribed pairs of interface arcs between disjoint partial objects or prescribed pieces. The rules maintain the interval and collar contract. They include all boundary identifications relevant to later stages; there are no unrecorded point or edge identifications.

- (iii) Within a fixed type and active set, the admissible complementary objects and arc maps are independent of the internal connectivity partition. Any required boundary framing, cyclic order, ambient region, boundary-chain constraint, matching condition, or parity label is included in the type or separately certified to be irrelevant to all future rules.
- (iv) Costs add under assembly. A terminal success is one disk with any prescribed external constraints encoded by the terminal type. A region decomposition, when ambient embedding matters, keeps interiors of pieces in disjoint regions and ensures that matching boundary types can actually be glued as embedded surfaces.
- (v) The specification is complete for its *own stated language*: every allowed assembly is generated by its finite rules. Completeness for all compressing discs in a manifold is an additional assertion, not part of this definition.

Here geometric admissibility means that the specified gluing and external constraints are well defined; it does not mean that the result is a disk independently of its partition. That latter question is still decided by D_r . Computational DAG sharing reuses descriptions, not an unrecorded identification of physical surface pieces.

Item (iii) is stronger than saying that two states have the same number of ports. For example, a transition allowed only when two specified arcs are on the same current component does not satisfy it unless that information has been included in the type. Adding enough type information can restore correctness while destroying a small type count. Both issues must be tracked.

Theorem 6.2 (Replacement is safe in every grammar context). *In a typed boundary-arc assembly grammar, replacing a family of live disk-union candidates of one type and active set by a minimum disk-representative subset preserves the minimum cost of every successful terminal assembly using that family. Repeated replacements at intermediate stages preserve the global optimum within the grammar.*

Proof. Fix a successful terminal assembly using a candidate S at the selected stage. Regard everything outside S in that assembly as one complementary surface T , with the already specified identifications among its own pieces performed. The type and collar contract says that its entire contact with S is the labelled arc interface. There are no hidden additional intersections. Components of S meet the interface by the live-state convention. Since the result is one disk and S is a proper nonterminal part, components of T also meet the interface. The complementary surface satisfies the surface-with-boundary contract.

Apply [theorem 2.1](#) to this final gluing. The complement T must be a union of disks, and its labelled arcs give a cap partition q . The chosen S satisfies $D_r(p_S, q) = 1$. By representativity, some retained candidate S' of the same type is completed by q at cost at most that of S . Item (iii) ensures the very same outside assembly and arc maps remain legal for S' , and item (iv) ensures that any ambient embedding and terminal type constraints are preserved. By the disk-completion criterion, their union is a disk. Additivity of cost gives a no-more-expensive successful assembly.

The retained family is a subset, so replacing it cannot introduce a cheaper assembly that did not already belong to the original language. Hence the optimum is unchanged. This argument applies to each replacement in a finite acyclic evaluation order, proving the iterated statement. $\square$

The proof uses the whole remaining context, not just the next local move. A retained candidate may react differently to an individual intermediate operation than a discarded one.

That causes no error when the type contract holds: the successful whole completion can still be substituted, and [corollary 2.2](#) ensures that its intermediate pieces will pass the disk-union and live-component tests. This is exactly why arbitrary history-dependent guards need separate treatment.

6.2 A complete sequential grammar implemented in this delivery

The implemented grammar starts with a nonempty finite family of disk unions on r_0 arcs. Layer i has an explicit list of options, each a disk union with r_{i-1} incoming arcs and $r_i \geq 1$ outgoing arcs. The incoming arcs are glued to the previous frontier by the fixed labels. A final family of disk-union caps consumes the last outgoing interface. Costs of the selected initial object, layer options, and cap are added.

At one transition, build the component graph of the current object and the selected layer option. If adding an edge creates a cycle, reject: positive defect is permanent. Otherwise each component of that forest gives a disk. Restrict its labels to the outgoing arcs; if any graph component has no outgoing arc, reject the transition because later material still has to be attached. Canonicalize the resulting output partition. Layer components that have only outgoing arcs are permitted: they are new disks that may be connected later. The graph includes them as isolated vertices until that happens.

After each layer, both the exact baseline and the reduced solver keep only the cheapest witness for an identical partition. The reduced solver additionally applies [theorem 4.1](#). At the cap, both evaluate (3) directly. By [theorem 6.2](#), both solvers return the same minimum cost or both find no assembly in the given grammar. A literal polygon-and-triangle replay checks a returned witness independently of the frontier transitions.

This is a complete search in an explicit finite language, not a simulation in which success has been assumed in advance. It is nevertheless an *abstract surface* language until actual embedded pieces and a source manifold are bound to it. The code accordingly returns `FOUND_ABSTRACT_DISK` or `NO_DISK_IN_GRAMMAR`. Neither is an unknot-recognition verdict.

6.3 End-to-end accounting within a supplied grammar

Theorem 6.3 (Single-exponential typed-grammar search). *Consider a supplied acyclic binary assembly grammar of explicit input bit length I , with H evaluation nodes, at most A geometric types per node, and at most $b \geq 1$ live arcs in each type. Assume at most K explicit local choices per ordered pair of child types. Let P bound the encoded work per local validation, state construction, integer cost operation, and witness-DAG update; also let P bound the full encoded size and verification cost of a required terminal witness. Then a deterministic minimum-cost disk-assembly search can be performed in*

$$2^{O(b)} \text{poly}(I, H, A, K, P) \tag{11}$$

bit time, with independently checkable pruning evidence and recoverable witnesses. The constants in the polynomial do not depend on b . Any cost of discovering or constructing this supplied grammar from a different input is additional.

Proof. Reduce each leaf family by type, charging its full explicit length and the output-sensitive bound (9). Summed over the leaves, the number of input candidates is at most I , so this contributes at most $2^{O(b)} \text{poly}(I, P)$. No Bell-sized list is generated implicitly.

Each child family has at most $A2^{b-1}$ candidates. A binary node therefore creates at most KA^24^{b-1} ordered combinations before local tests. Collect successful ones by output type and

active interface, and reduce each collection. Formula (9), with $d \leq 2^{b-1}$, gives a conservative factor at most another 4^{b-1} , up to the polynomial input, cost, and sorting factors. Thus a bound of the form

$$O\left(HKA^2 16^b \text{poly}(b, P, \log(HKA + 2))\right)$$

is sufficient for the internal nodes. We do not claim this base is optimized. If varying active sets are not already counted as types, [corollary 5.1](#) replaces the per-type bound by $O(3^b)$; the same argument remains single exponential. Apply [theorem 6.2](#) after each reduction. Finally check terminal records and replay a minimum witness. The parameter P includes any required expanded witness size, not merely its DAG size. $\square$

The input-length parameter is essential. A binary-compressed grammar description whose explicit expansion is huge is not covered by pretending that expansion is free. Similarly, small representative tables do not bound the discovery cost of new geometric types or the cost of expanding a shared witness into an actual surface certificate.

Corollary 6.4 (Polynomial and quasi-polynomial regimes). *Suppose an n -crossing input has a complete geometric disk-search grammar satisfying [definition 6.1](#), constructible with polynomial encoded overhead, polynomially many types and rules, and polynomial-size witnesses and local verification. If its live-arc bound is $b = O(\log(n + 2))$, the search above is polynomial. If $b = O((\log(n + 2))^2)$, it runs in*

$$\exp(O((\log(n + 2))^2)) = n^{O(\log n)}$$

for $n \geq 2$. If terminal success is a properly embedded essential-boundary disk in a source-certified knot exterior and the grammar is complete for those disks, this yields a recognition algorithm within the same bound.

Proof. Substitute the bounds into [theorem 6.3](#). The last assertion additionally uses the compressing-disc characterization discussed in [section 7](#), and the assumed completeness of the grammar. $\square$

The corollary isolates a genuine improvement in the allowable frontier scale. Direct Bell-state enumeration pays $2^{\Theta(b \log b)}$, giving the sufficient scale $b = O((\log n)^2 / \log \log n)$ for a similar quasi-polynomial bound. Here the $\log b$ loss disappears for the disk-completion language. This comparison does not imply that the predecessor's more general coning language can be reduced by the same rule. Nor have we proved the stated width or completeness assumptions for all knot diagrams.

7 Connecting the kernel to the native compressing-disc route

7.1 What the reviewed native modules already provide

The maintained synthesis [\[7\]](#) describes three relevant entry points:

```
normal_surface_components.normal_component_census
normal_component_verify.verify_normal_component_certificate
normal_disk_kernel.normal_compressing_disk_count.
```

Their purpose is to analyze a *supplied* admissible normal vector in a validated compact triangulation. Binary multiplicities remain compressed. The source-checked exterior construction [\[6\]](#) binds one canonical triangulation to a validated input knot diagram. Those facts make a

constructive disk search more useful than an isolated partition calculation, but they do not supply the assembly grammar assumed in [corollary 6.4](#).

The native essential-disc criterion is componentwise: a connected component with Euler characteristic one and a nonzero pair of torus-boundary parity evaluations is a compressing disk. A closed projective plane also has Euler characteristic one but has zero boundary values, so it is excluded. Taking only the total boundary parity of a whole normal surface is insufficient; parallel meridian disks can cancel in the aggregate. Our search instead assembles a single disk and must retain its own boundary class. This is an application of the existing criterion, not a replacement for the native component census.

7.2 Two homology bits do not destroy the state bound

Let M be a source-certified knot exterior, with boundary torus $\mathbb{T}$. Fix a certified basis of $H^1(\mathbb{T}; \mathbb{F}_2)$. Suppose a geometric grammar assigns to each piece its external boundary chain on $\mathbb{T}$ and ensures that all internal arc identifications cancel the appropriate chain contributions. The two cocycle evaluations of the final boundary are then the sum of the piece contributions, including any certified local correction terms. This is a *chain-level* requirement: naming two arbitrary stored bits “homology” does not meet it.

Proposition 7.1 (Essential-boundary refinement). *Under the preceding certified chain contract, reducing separately for the four possible boundary parity labels preserves minimum-cost disk completions with any prescribed final parity. It uses at most $4 \cdot 2^{r-1}$ representatives per other fixed type and active interface. A terminal properly embedded disk with nonzero parity is a compressing disk in M .*

Proof. For a fixed complementary object, the required partial parity is determined by the specified final parity and the complementary contribution. The representative theorem applies within that parity class, so a no-more-expensive replacement leaves the final parity unchanged. There are four classes.

The boundary of a properly embedded disk is an embedded circle in the torus. An inessential embedded circle has zero homology class. An essential embedded circle on a torus represents a primitive integral class (a, b) ; a primitive pair cannot have both entries even, and hence has nonzero reduction modulo two. Thus nonzero parity is precisely essentiality for this circle. The disk is therefore a compressing disk. $\square$

For a genuine knot exterior in S^3 , the existence of a compressing disk for its boundary torus characterizes the unknot: the exterior is irreducible, boundary compression gives the solid-torus case, and the knot is unknotted. This is the geometric endpoint already used in the native source theorem [\[6\]](#). The source binding matters. A disk in an unrelated solid torus says nothing about the input knot.

7.3 The integration gate

A native use of this reducer needs four independently identifiable layers. First, a producer must export actual disk pieces, their labelled boundary intervals, and their embedded regions or other sufficient compatibility types from the current geometric representation. Second, the search must account for all rule generation and all type and width growth. Third, each rank reduction must be checked against the actual family it prunes. Finally, an accepted assembly must replay to a properly embedded essential-boundary disk in the *same* source-certified exterior.

The package supplies the third layer and an independent *abstract-surface* form of the final replay, together with a complete solver for its supplied finite grammar. It does not supply the first layer, a complete normal-vector search, a general hierarchy compiler, or a native embedded-disk certificate exporter. The canonical exterior producer itself is *not* missing at the reviewed pin. The remaining gap is the complete, type-controlled disk-assembly search attached to it.

The intended integration is an isolated experimental report or library module. No maintained dispatch should infer **KNOTTED** from the absence of a disk in an incomplete grammar. A successful abstract polygon gluing should not infer **UNKNOT** without its ambient embedding and source certificate. Resource-limit exceptions are inconclusive and must preserve access to the existing exact fallback.

8 Safety boundaries and explicit counterexamples

8.1 A same-component guard is not preserved

Take the three candidates of [example 3.4](#) with costs 0,0,1, in the order 001,010,011. The reducer keeps the first two and discards the third, because its root vector is their sum. It still preserves the minimum cost for *every* disk-completion query.

However, the predicate “labels 1 and 2 lie in the same block” holds only for 011 among these candidates. A subsequent rule enabled only by that predicate would lose its only admissible source. Therefore the reducer is not a valid optimization of arbitrary partition-dependent transition systems. This is an executable negative test in the delivery. The correct repair is to prove an appropriate context contract or split types by the relevant predicate, and then count the extra types.

In particular, homogeneity guards for arbitrary interval attachments in the preceding port report may depend on component information not preserved here. Neither the Bell-state lower bound nor that report’s guard semantics is contradicted by our result. We preserve a narrower terminal language rather than reconstruct the entire state.

8.2 Compressed point incidence is not geometric arc incidence

The sparse component census counts components meeting specified point sets. One row can describe many components and a marked interval can represent many normal arcs. Our width r counts actual boundary intervals in a surface gluing. The example of two disks glued along two arcs, rather than one, proves that simply using the number of histogram rows or named point ports as r is unsound.

A future compressed version could attach a binary multiplicity to a family of parallel arcs, but it would need to track the resulting Euler correction, attachment cycles, possible orientation effects, and geometric compatibility. The present theorem does not turn an exponential number of actual arcs into a logarithmic interface by naming their bundle once.

8.3 Boundary constraints, hidden types, and circles

A partition does not encode every ambient embedding constraint. Different cyclic orders, different positions on an incompressible boundary pattern, or different quadrilateral choices can be incompatible with the same future region. The type contract must expose whatever data the continuation can inspect. If there are $2^{\Theta(b \log b)}$ such types, a 2^b representative count

inside each type has not eliminated the overall Bell-scale obstruction.

Gluing along a whole boundary circle is also not an arc gluing. Its Euler correction is zero rather than one, and capping by a disk can remove boundary in a way excluded by the defect monotonicity argument. Likewise, compressing an annulus by an actual surgery is not an attachment of surfaces along disjoint boundary intervals. The persistence theorem must not be extended to these operations without a new proof.

8.4 Optimization is not counting, and certificates have scope

A representative subset preserves minima and existence for the specified completion predicates. It does not preserve the number of successful candidates, their distribution of witness topologies beyond the recorded type, or a full Pareto frontier of several costs. Linear dependencies may eliminate many distinct valid witnesses. This is intentional for decision and scalar optimization, but it is unsuitable as a counting algorithm without additional machinery.

The algebraic checker certifies source-bound row identities and cost dominance. The PL replay certifies the topology of the returned abstract glued surface. Neither checks an embedding in a 3-manifold, normal matching equations, diagram-to-exterior provenance, or the completeness of a producer's rule set. Those are separate proof obligations, even when all local code checks pass.

9 Implementation, validation, and paired measurements

9.1 The independent triangulated-surface oracle

For each partition block containing k labels, `code/surface_model.py` constructs a polygonal disk with $4k$ boundary vertices, triangulated from a new central vertex. It marks k separated boundary edges as the attachment arcs, leaving gaps between their endpoints. For two frontiers or a whole grammar witness, it glues the selected boundary edges with the prescribed endpoint maps.

The oracle then constructs the quotient simplicial complex. It checks triangle and edge incidence and checks every vertex link is a connected path or circle, thereby rejecting nonmanifold identifications instead of merely counting cells. It computes connected components, Euler characteristic, boundary-circle count, and orientability by propagating triangle orientations. These computations do not call the root-vector or transition routines. The result is a genuine finite surface calculation, not the partition predicate restated as an expected value.

This oracle intentionally expands small polygonal fixtures. It is not a compressed normal-surface algorithm, and its triangle count is part of its runtime. An abstract surface can be checked this way without claiming to know where it embeds in a knot exterior.

9.2 Executed validation and its exact size

The retained environment was CPython 3.13.5 on Linux x86_64 with glibc 2.41. The standard-library unit suite contains **36 test methods**, all passing. These methods include exhaustive factorization checks through $r = 6$, root-feature graded ranks through $r = 8$, independent row construction, signed all-pair checks through $r = 5$, width and cancellation controls, huge 4096-bit signed costs, source and certificate mutations, literal small-surface gluings, and complete layered searches. Loop iterations inside a test are not additional test methods.

The separate audit, seeded by 20261009, records:

Audit operation	Completed checks
Entries of direct completion matrices, $r = 1, \dots, 7$	813,297
Weighted candidate families and certificate replay	1,000
Minimum-completion queries on those families	14,014
Random literal triangulated-surface gluings, $r \leq 8$	1,000
Complete variable-width grammar comparisons	200

The direct matrix ranks are computed from the independently constructed graph predicate, *not* by taking the rank of the proposed feature vectors. They are 1, 2, 4, 8, 16, 32, 64 for widths 1 through 7, as predicted. The weighted audit checks both preservation of finite minima and preservation of infeasibility. The complete-grammar audit compares reduced and unreduced searches, checks each retained pruning certificate, and replays every successful witness as a triangulated surface.

Negative controls include the two-arc annulus, a twisted Möbius-band gluing, a same-component guard lost after otherwise valid reduction, rejection of sealed components, and persistence of an annular defect after a further disk attachment. The latter replay has Euler characteristic zero and two boundary circles, not a disk. All audit checks passed. These finite checks support implementation reliability but do not replace the universal proofs or constitute proof-assistant verification.

9.3 Benchmark design and fair baseline

The benchmark is a complete *explicit abstract assembly grammar*, not a collection of knots. For each $r = 6, 7, 8$, it supplies all partitions as an initial candidate family with deterministic seeded integer costs. It then supplies two layers. Each layer has an identity option and an option for merging each pair of strands, with explicit costs, followed by a fixed cap. Both solvers use exactly the same input, validation, local transition generator, exact duplicate-state elimination, and witness bookkeeping. The only algorithmic difference is representative reduction between stages.

Each mode is run three times with alternating run order; the table reports medians. Input grammar construction and the final literal replay are excluded from both solver timings. Validation, candidate deduplication, reduction, cost arithmetic, and witness handling inside the solver are included. The initial list is Bell-sized by design, and both modes must read it. These measurements therefore demonstrate amortization across later search layers, not an algorithm for generating a Bell-sized family in single-exponential time. The input language is deliberately transparent; specialized algorithms for a particular generated family may improve on both generic solvers.

r	Initial states	Reduced states	Exact (s)	Reduced (s)	Ratio
6	203	32	0.02376	0.00598	3.98
7	877	64	0.16647	0.02907	5.73
8	4,140	128	1.38119	0.11594	11.91

Table 1: Paired complete abstract-grammar search. The ratio is exact median time divided by reduced median time. Reduced-state counts are the stored counts at each of the three stages on these inputs. These are not native recognizer speedups.

At width eight, the unreduced search attempts 240,120 layer transitions; the reduced search attempts 7,424. It stores 128 states instead of 4,140. The corresponding median speed ratio is 11.91, less than the transition-count ratio because reduction and other bookkeeping still cost

r	Exact transitions	Reduced transitions	Optimum cost	Replay triangles
6	6,496	1,024	-82	144
7	38,588	2,816	-94	168
8	240,120	7,424	-97	192

Table 2: Deterministic work counters for the same experiments. The two solvers agree on the optimum. Every returned reduced witness replays to one orientable surface with Euler characteristic one and one boundary circle.

time. Before the second and third reductions, its deduplicated candidate counts are 1,072 and 943; pre-reduction work has not been hidden behind the final 128-state bound.

9.4 A negative timing result: do not compress a lone query by default

A width-eight control uses the same initial family and cap but no intermediate layer. The direct solver takes 0.01397 seconds, while the representative version takes 0.06553 seconds, approximately 4.69 times longer. Both return cost -97 . This control is a single recorded timing per mode, not a three-repeat median. Its purpose is to show a real overhead case, not to estimate stable hardware performance from one sample.

A single cap can be checked directly in $\text{poly}(r)$ work per candidate. Constructing and reducing 2^{r-1} -bit vectors before that one query is often wasteful. The benefit appears when enough subsequent candidate generation or cap queries reuse the smaller family. An eventual native scheduler should retain a direct path and compare amortized savings against construction and checking costs. The present measurements justify neither enabling this kernel everywhere nor advertising a whole-recognizer speed factor.

9.5 What was not run

The maintained ProveIt test suite was not executed in this environment. No patch was applied to `fast/`, and no native normal-vector, hierarchy, or knot-corpus benchmark was performed. The repository sources and incoming directory were inspected through the available read interfaces; the incoming binary archive collection was not fully unpacked and audited. The predecessor port report was reviewed through its available Library text. The source-audit file records this coverage rather than claiming an exhaustive review of every incoming manuscript.

10 Further research questions and concrete next milestones

10.1 Geometric construction and completeness

1. A native arc-interface exporter. Can the current source-checked compact exterior and normal-component modules export a supplied disk decomposition with genuine separated boundary arcs, explicit collars, and an independently replayable embedding? The first target is correctness on supplied witnesses, not discovery. A successful exporter should distinguish point-incidence ports from the arc labels used in this article and verify all chain-level boundary parities.

2. A complete disk-assembly grammar. Which normal or hierarchy decompositions guarantee that every essential disk has an assembly generated by polynomially described local

rules? The proof must show that intermediate cuts satisfy the boundary-arc contract, that no omitted surgery is needed, and that every required normal-coordinate choice can actually be generated within the claimed bound. A small successful witness is not a bound on the complete search.

3. Controlling geometric types. Can cyclic order, normal matching, quadrilateral compatibility, ambient region data, and external-boundary constraints be organized into only $2^{O(b)} \text{poly}(n)$ types? If a stronger polynomial type bound is false, isolate the smallest datum producing super-single-exponential growth. This is as important as reducing partition counts.

4. A width theorem tied to a hierarchy. Does a complete compressed hierarchy admit disk-search frontiers with $b = O((\log n)^2)$ and polynomially bounded witness encodings? A width estimate for a projection graph alone does not bound the number of actual surface arcs carried through that frontier. A proof needs a global encoded potential paying for arc births, refinement, coordinate growth, and all visited geometric types.

10.2 Sharper algebra and compressed geometry

5. Multiplicity-aware arc bundles. Find a compact completion algebra for many parallel boundary arcs that preserves the cycle defect and orientation data. A desirable theorem would use logarithms of binary multiplicities rather than expanding all arcs, while still reproducing the annulus and Möbius negative controls. The one-port surrogate without correction is ruled out by [section 8](#).

6. Faster graded reduction and joins. The implementation uses conservative dense bit vectors. Can the rooted-forest minors be streamed, factored, or kept in sparse graded form to avoid d^2 storage in common cases? Representative product-family methods [\[3\]](#) suggest reducing joins without first constructing every child pair. Any improved bound should include certificate generation, integer weight comparisons, and witness recovery, not only field-operation counts.

7. Persistent and changing-interface families. Can verified bases be reused under rollback, changes of costs, or insertion and deletion of active labels? Cost monotonicity in the present certificate is tied to one input ordering, and arbitrary future weight changes invalidate it. A useful persistent scheme should maintain checkable source binding and charge its work across the actual search DAG, not just one successful branch.

8. Bounded-defect completion. For targets of genus at most g or a prescribed boundary pattern, determine the rank of the completion predicate after simultaneously recording defect, component count, and the signed consistency data of [theorem 5.2](#). The disk case has the special zero-defect obstruction. Simply accepting connected signed states does not control annuli or handles. It would be useful to identify parameter ranges admitting $2^{O(b+g)}$ rather than Bell-scale kernels.

10.3 Verification, scheduling, and limitations

9. Multiobjective costs and actual geometric weight. Normal weight, encoded complexity, boundary data, and future width may be competing objectives. Does a useful restricted Pareto order permit small representative families, or do unavoidable trade-offs force

many representatives? For scalar cost, determine which geometric weights remain exactly additive under the chosen local rules and which need correction terms.

10. Whole-pipeline ablation. After the native exporter exists, benchmark complete recognition with and without representative reduction on the same knot inputs and budgets. Include direct cap queries, already-small families, failed grammar searches, and large-type cases. Record candidate generation, geometry validation, certificate replay, and fallback work separately. The negative one-query control should remain a required regression.

11. Formal proof and checker hardening. A compact formalization can begin with the defect identity, the rooted-forest minor lemma, the exact matrix factorization, and cost-dominating row expansions. The geometric type/context replacement theorem is a separate formalization task, not a consequence of verifying Gaussian elimination. On the implementation side, bounded allocation, adversarial certificate size, serialization, and cancellation throughout a long grammar evaluation need production-level engineering.

12. Sharp families inside knot geometry. The coordinate partitions prove optimality in the abstract interface model. Which of them can coexist with one common boundary type in a natural decomposition of a knot exterior? A realizability theorem would clarify whether the sharp bound is relevant in practice; an obstruction could yield a still smaller kernel. Conversely, construct families where the partition kernel is small but type growth or actual arc width defeats the desired global bound. Such examples would identify the true next bottleneck.

11 Conclusion

Bell-many distinguishable physical states do not force Bell-many representatives for minimum disk completion. The reason is not that topology can be ignored. The exact defect identity first shows which topology is relevant: all pieces must be disks and their attachment graph must be a tree. That specific predicate has an explicit root-transversal factorization, exact rank 2^{r-1} , and sharp graded representative bounds. Weighted reduction retains actual witnesses and admits source-bound independent checking. The typed replacement theorem then justifies reuse inside a finite assembly search, rather than merely optimizing an isolated matrix calculation.

The verified standalone implementation and its complete-grammar comparisons demonstrate both the potential benefit and the overhead. The current ProveIt exterior and component certificates provide a concrete destination for the next geometric integration, but the complete bounded-width disk-assembly producer is still an unproved obligation. The present result removes a Bell-number factor from a precisely specified search layer; it does not disguise the remaining search and geometry costs as constant-time operations.

A Reproduction and artifact map

From the extracted package directory, run:

```
make test
make audit
make example
make benchmark    # new timings; overwrites results/benchmark.json
make tables       # reads retained JSON only
make pdf          # requires pdflatex; no benchmark rerun
```

The Python code requires only the standard library and uses syntax compatible with Python 3.10 or later; the executed environment is recorded above. The PDF uses the LaTeX packages listed in its preamble. The recorded timing results are retained so that rebuilding the article does not silently rerun the experiments.

File	Role
article.tex, article.pdf references.bib	Self-contained article and compiled delivery. Bibliographic export; the TeX also contains its own reference list.
code/disk_kernel.py	Root vectors, direct completion, cost-ordered representative subset, and expansion certificates.
code/certificate_check.py	Separate row reconstruction, graph predicate, strict algebraic certificate verification.
code/assembly.py	Exact and reduced finite layered-grammar solvers; literal witness replay entry point.
code/surface_model.py	Polygon triangulation, gluing, manifold checks, Euler characteristic, boundary and orientation replay.
code/signed_kernel.py	Signed root-coloring and direct compatibility kernels.
tests/test_research.py	The 36-method standard-library unit suite.
code/audit.py,	Deterministic mathematical audit and paired
code/benchmark.py	complete-grammar measurements.
code/example.py	A small certificate example and a full grammar witness with literal surface replay.
results/	Executed logs, raw JSON measurements, and generated table fragments.
docs/CLAIMS.md	Proof, implementation, test, and integration status by claim.
docs/INTEGRATION.md	Safe interfaces and explicit native integration gates.
docs/SOURCE_AUDIT.md	Reviewed revision, source coverage, literature, and novelty limitations.

B A complete small reduction example

```

from disk_kernel import Candidate, reduce_family
from certificate_check import verify_certificate

items = [Candidate((0, 0, 1), 0),
          Candidate((0, 1, 0), 0),
          Candidate((0, 1, 1), 1)]
answer = reduce_family(items)
assert len(answer.retained) == 2
assert verify_certificate([x.partition for x in items],
                          [x.cost for x in items],
                          answer.certificate)
# For the third row the expansion is [0, 1].
# This certifies every disk-completion optimum, not arbitrary guards.

```

The default root-vector cap is 20 labels. It is an allocation guard, not a mathematical bound on the theorem. The layered prototype also validates the combined incoming and

outgoing partition of each piece under its current width guard. Its intended role is small exact experimentation and independent replay, not an uncapped production engine. The reducer accepts a cooperative callback; an exception propagates rather than publishing a partial reduction as a complete result. The top-level grammar solver is finite but does not yet implement a global cooperative cancellation budget. A native integration must add and test that resource contract.

C Source and authorship note

This package was prepared against the pinned repository listed in [section 1](#). The incoming intake policy assigns the next report number during placement and retains a delivered report separately from maintained runtime changes. Accordingly, this package does not choose a report number, modify the upstream repository, or include an automatically applied patch. Its provenance document is a source-and-scope record, not a checksum-only manifest. No third-party font files, upstream source archive, or copied research-paper PDF is redistributed.

The mathematical assertions are human-readable proofs and finite computational checks. They have not been externally peer reviewed or formalized in Lean or another proof assistant. Established rank-based and matroid-representative methods are explicitly credited, and the global unknot-recognition corollary remains conditional on its stated geometric construction hypotheses.

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